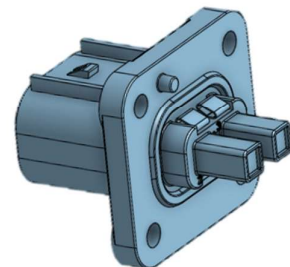
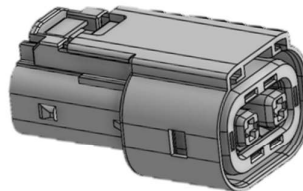
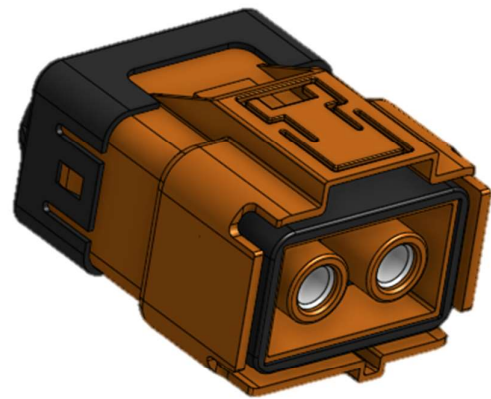
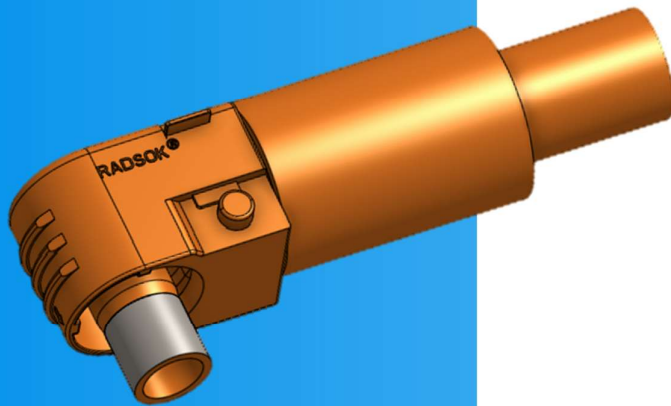




How To Crimp NU25 HV Connectors

ZERO-TO-HERO



Jacob Lukes
2025





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Sanity Checks

- Ensure all components needed for crimping the connectors are present
- Make double sure the crimp needs to happen
- Ensure correct crimp orientation of connectors

HIP to Motor Controller

SLPPB50BSO0EH / SLPPB50BSO1EH

- **SLPPB50BSO0EH** is a **90-degree** keyway connector
- **SLPPB50BSO1EH** is a **270-degree** keyway connector
- Ensure the following components are present (see Figure 1, part A)
 - Connector
 - Inner copper adaptor
 - Not glue-lined Heat-shrink
 - Glue-lined heat-shrink
 - Rubber gromet
 - Plastic cable ring
 - Cable retainer
- Step 0
 - Check rotation of connector when it is crimped to cable (permanent marker is suggested to get this right)
- Step 1
 - Thread on components in this order (see Figure 1, part B):
 - Cable retainer (ensure facing correct direction)
 - Plastic cable ring
 - Rubber gromet
 - Not glue-lined heat-shrink
 - Glue-lined heat-shrink

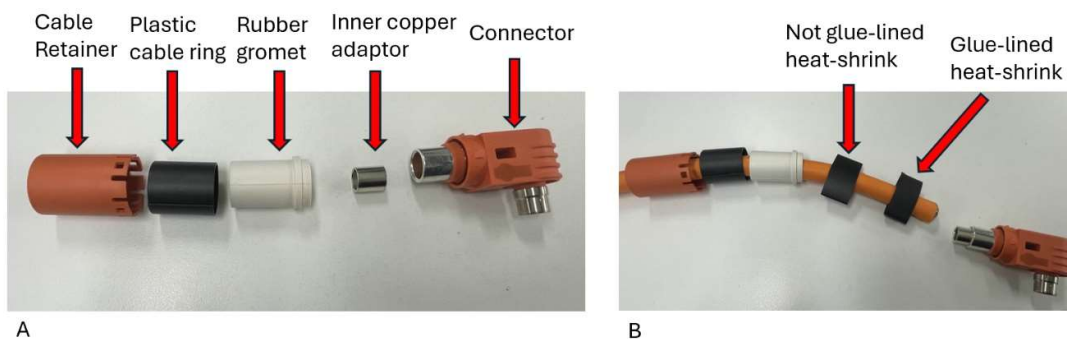


Figure 1

- Step 2
 - Strip back 28 mm of outer insulation, using either a knife (refer to: Stanely Knife) or a cable stripping tool
 - Cut off and shorten outer foil shield down to the end of the outer insulation
 - Carefully and slowly unbraid wire shielding
 - Once unbraided, use scissors (refer to: Scissors) to shorten back the wire shielding back to the original 28 mm length (see Figure 2)



Figure 2

- Step 3
 - Using cable shears (refer to: Cable Shears), cut off 10 mm of the inner copper and insulation (as seen in Figure 2 and Figure 3)



Figure 3

- Step 4
 - Curl back the wire shielding
 - Strip back 15 mm of the inner copper core insulation, using a Stanley Knife
 - Insert inner copper adaptor
 - See Figure 4



Figure 4

- Step 5
 - Insert and crimp the connector to the copper, after the first crimp, rotate $\sim 45^\circ$ to ensure strong crimp
 - Use the big Blue Handle Cable , on the 95 mm setting, see Figure 5
 - Figure 6 shows what happens if 70 mm setting is used



Figure 5

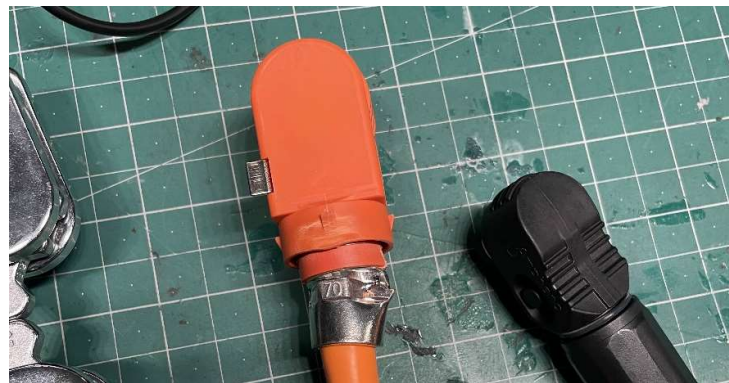


Figure 6

- Step 6
 - Thread the glue-lined heat-shrink over the wire shielding to cover the crimp
 - See Figure 7, part A
 - **Ensure that the crimp and the shielding are not touching**
 - If they do touch after the heat shrink is shrunk down, then when it goes on the car there will be an IMD fault, due to HV and LV ground touching. You'll be chasing your tail for hours
 - Heat-shrink it down
 - Heat gun at ~250 °C, with second or third flow rate setting
 - Figure 7 part B and C show how the crimp and shield wires are not touching
 - The outer insulation and shielding may need to be pulled down to ensure that the wire shield and crimp are not touching

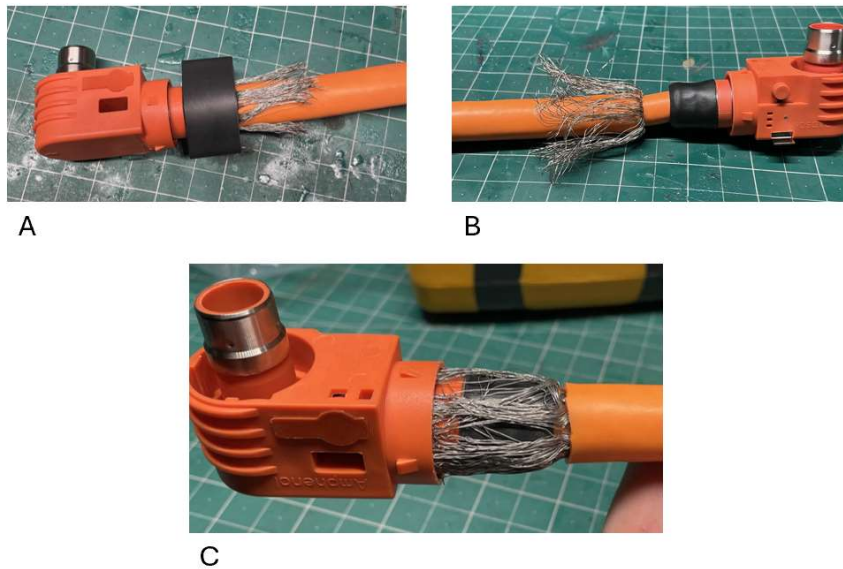


Figure 7

- Step 7
 - Bring up and shrink down the not glue-lined heat-shrink, to cover and secure the wire shielding, refer to Figure 8

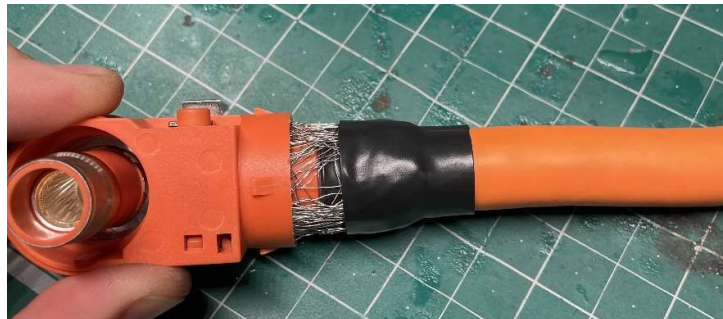


Figure 8

- Step 8
 - Bring up rubber gromet, see Figure 9
 - You may need to cut and comprise this gromet, depending on how Step 10 goes



Figure 9

- Step 9
 - Bring up plastic cable ring to cover rubber gromet

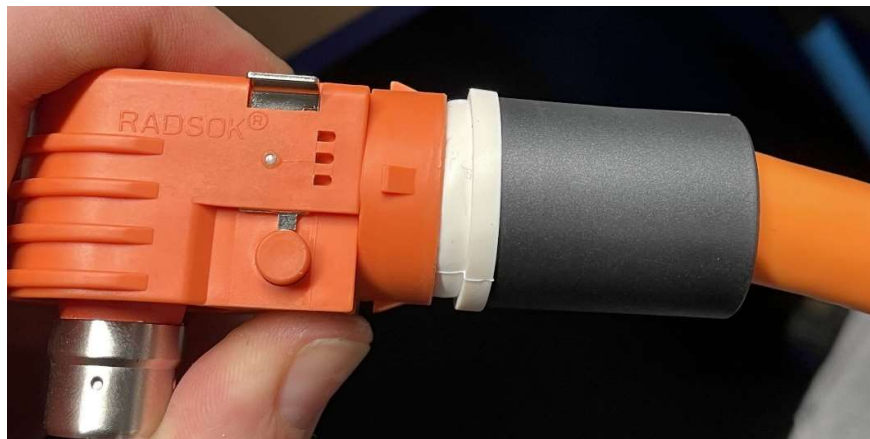


Figure 10

- Step 10
 - Bring up the cable retainer, ensure that it comes up far enough for the clips to click in (refer to Figure 11 for the clips in question)

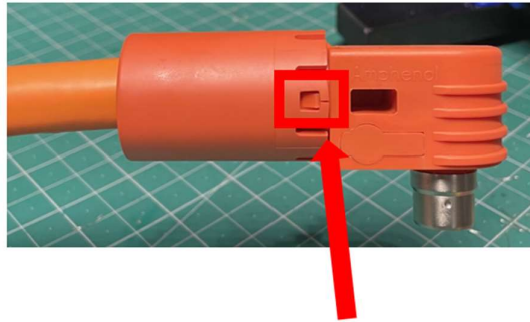


Figure 11

- This part will most likely take a lot of force, if there is no hope of it working, go back to Step 8 and cut/'comprise' the gromet, so that when the cable retainer is pushed up, the gromet will break inside of it. This should allow the retainer to easily clip up
- See Figure 12 for the completed assembly

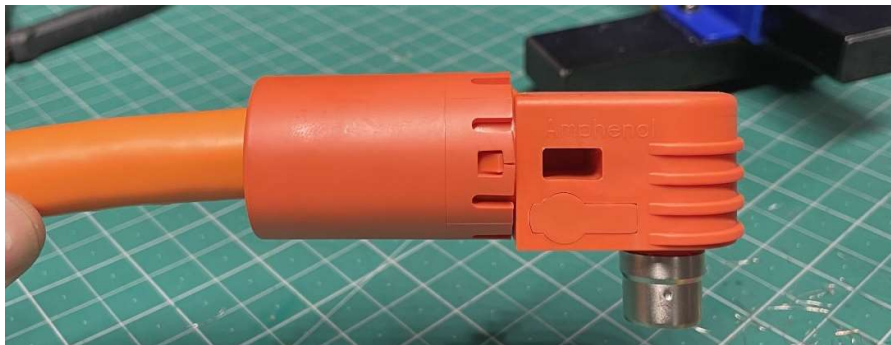


Figure 12

HIP to DCDC/Energy Meter

2103124-4

- Assembly instructions found in “2103124-4 Assembly Instructions” pdf available on the NU Teams SharePoint shown in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

Documents > NU Racing > Electric Vehicle > 2025 > MCHA > HV Connectors > HIP to DCDC and Energy Meter

Name	Modified	Modified By	Part Number	+ Add column
2103124-4	July 8, 2025	Jacob Lukes		
4-2103177-4	July 8, 2025	Jacob Lukes		
HV Rating compliance	July 8, 2025	Jacob Lukes		
2103124-4 Assembly Instructions.pdf	July 8, 2025	Jacob Lukes		

4-2103177-4

- Assembly instructions found in “4-2103177-4 Assembly Instructions” pdf available on the NU Teams SharePoint shown in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

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2103124-4	July 8, 2025	Jacob Lukes		
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HV Rating compliance	July 8, 2025	Jacob Lukes		
2103124-4 Assembly Instructions.pdf	July 8, 2025	Jacob Lukes		
4-2103177-4 and 2103124-4 Required Components.txt	July 8, 2025	Jacob Lukes		
4-2103177-4 Assembly Instructions - HV RATING.pdf	July 8, 2025	Jacob Lukes		

HIP to Accumulator

ELPA2A2510

- Use 25 mm² on Blue Handle Cable Crimpers
- Make sure copper is all the way up to the hole on the pin



- Rotate crimp when crimping



- Use 25 mm² setting



- Further instructions can be found in the “ELPA2A2510 & ELRA2A0301 Assembly Instructions” pdf available in the folder shown below. (9/01/2026: I meant to append this document at the end of this one, but do not have time to do this before my report is due 😞)

Documents > NU Racing > Electric Vehicle > 2025 > MCHA > HV Connectors > HIP to Accum

 Name	Modified	Modified By	Part Number	+ Add column
 ELPA2A2510	July 8, 2025	Jacob Lukes		
 ELRA2A0301	July 8, 2025	Jacob Lukes		
  ELPA2A2510 & ELRA2A0301 Assembly Instructions.pdf	...	A few seconds ago	Jacob Lukes	
 epower-lite-series-ib-13.pdf	July 8, 2025	Jacob Lukes		

Tools

Scissors



Figure 13 Scissors

Cable Shears



Figure 14 Cable Shears

Blue Handle Cable Crimpers



Figure 15 Blue Handle Cable Crimpers

Stanley Knife



Figure 16 Stanely Knife